AI Customer Support Platform

PRODUCT OVERVIEW

An AI-native customer support platform

Text and voice conversations, answers grounded in your knowledge, authorized actions, and a human whenever it matters.

→ next chapter · ↓ details within a chapter · Esc overview · S speaker notes

CHAPTER 1

Vision and users

What we are building, for whom, and the principles we will not trade away.

VISION

Complete conversations, not just chatbot replies

Conduct complete, context-aware customer conversations over **text and voice**, answer from **trusted business knowledge**, perform **explicitly authorized actions**, and **transfer to humans** when appropriate.

Faster correct answers

Reduce time to a correct answer for repetitive requests.

Humans stay one step away

Resolve routine work without ever removing access to people.

Auditable AI

Every action is permission-controlled and recorded.

USERS

Five kinds of people

Customer

Asks questions or requests an action.

Support agent

Reviews, takes over, resolves.

Support lead

Queues, policies, reporting, quality.

Org admin

Channels, knowledge, agents, integrations, access.

Platform operator

Reliability, cost, abuse, health.

PRINCIPLES

Eight principles

Conversation first
One durable record; channels only deliver

Same brain for text and voice
One orchestration and authorization model

Grounded answers
Approved knowledge, honest uncertainty

Safe actions
Models propose, application code authorizes

Human escape hatch
Customer-requested or policy-driven

Tenant isolation by default
Data, search, tools, events, analytics

Observable and testable
Decisions, tools, handoffs, failures measured

Provider portability
Models, speech, channels behind adapters

HOW WE WILL KNOW

Success measures

- Correct-answer rate on a curated evaluation set, and groundedness
- Task completion rate, human escalation rate segmented by reason
- Median and p95 response latency, tool success rate
- Customer satisfaction and cost per resolved conversation
- Availability and message-delivery success

CHAPTER 2

The customer experience

One conversation that can switch between text, voice and a person.

A CONVERSATION

From question to resolution

1. **Ask**
   Website chat today, voice and email later
2. **Understand**
   Context from the conversation and customer
3. **Answer**
   From approved knowledge, with sources
4. **Act**
   Authorized lookup or action, confirmed by the tool
5. **Escalate**
   A person takes over with full context

History survives channel changes where identity can be safely linked.

USE CASES

What each person can do

Customer

- Ask a question
- Check an order
- Ask for a human
- Talk by voice

Support agent

- See the queue
- Take over with context
- Resolve, create tickets

Admin

- Configure agent and channels
- Upload knowledge
- Manage users and tools

Lead and operator

- Routing and SLA
- Quality and reports
- Health, cost, abuse

SCOPE

First release and what waits

In the MVP (text)

- Organization and access foundation
- Website text chat
- Knowledge ingestion and retrieval
- AI orchestration with grounded answers
- One read-only business tool
- Human handoff and inbox, basic tickets
- Audit, logs, metrics, traces, usage

Later

- Real-time browser voice (STT/TTS or speech-to-speech)
- Email and messaging channels
- Workflow builder, routing, SLA
- More connectors and write-capable tools
- Advanced analytics, evaluation, billing, governance

BOUNDARIES

Explicit non-goals for the first milestone

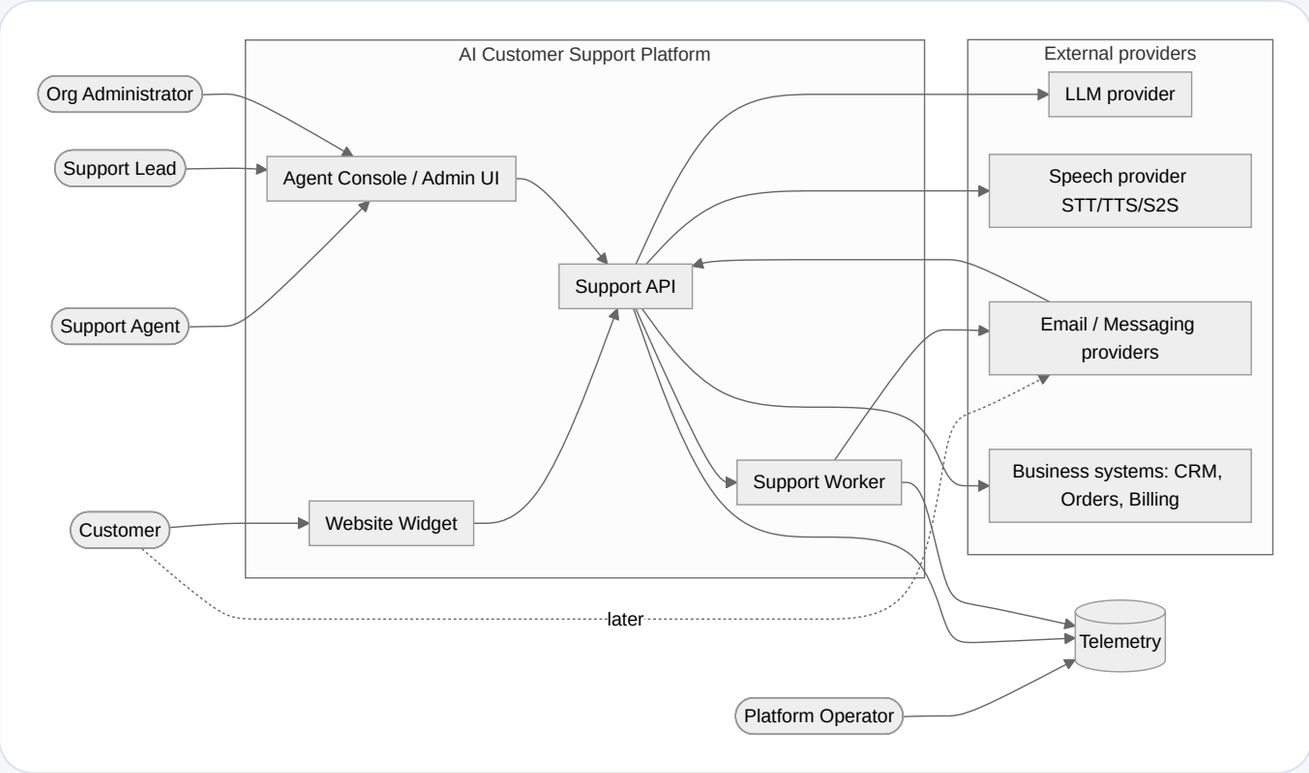
- ✕ Replacing every existing CRM or helpdesk
- ✕ Arbitrary autonomous changes to customer accounts
- ✕ Unreviewed financial transactions
- ✕ Supporting every channel and provider at launch
- ✕ Training a foundation model
- ✕ Splitting every module into a microservice

CHAPTER 3

Architecture

A modular monolith with strict module boundaries, ready to split only when it pays off.

The system in context

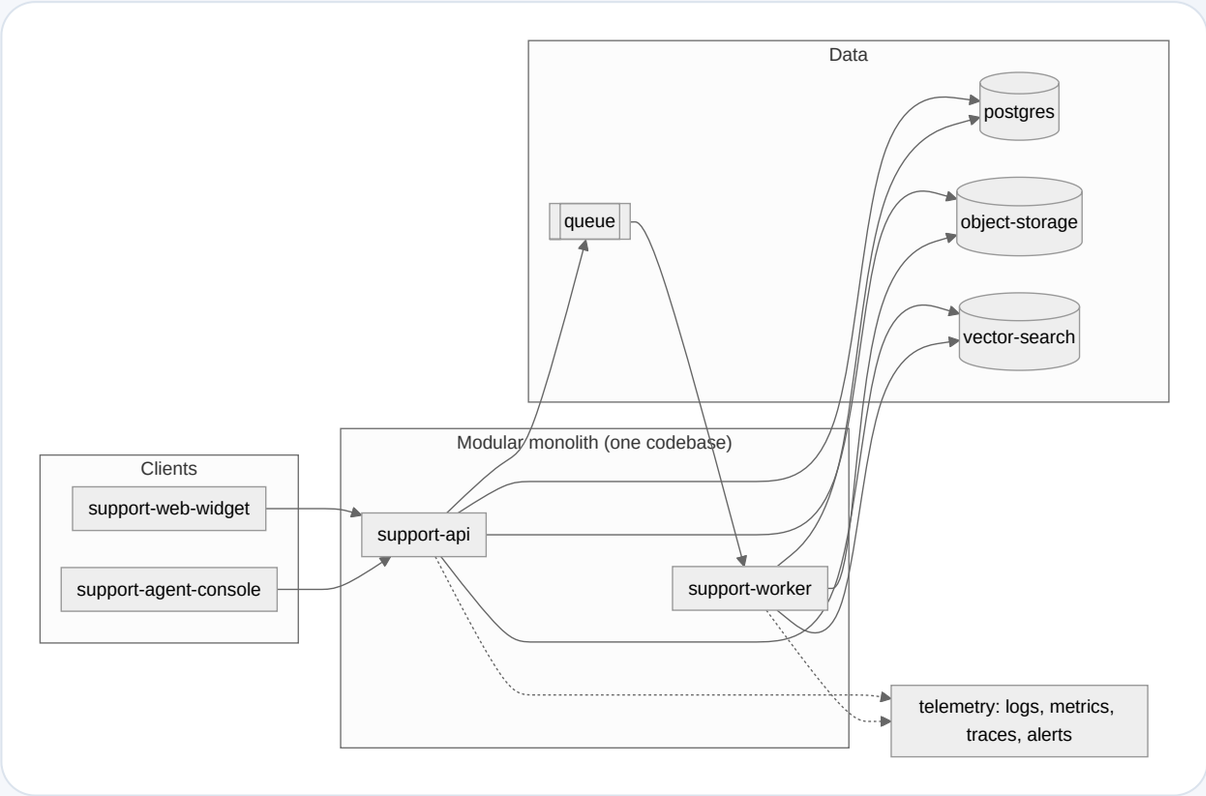


People on the left, providers on the right, one platform in the middle

RECOMMENDED START

Modular monolith first

- One deployment, strict module packages and interfaces
- Relational database for transactional state
- Queue and worker for retryable, slow work
- Vector search and object storage behind adapters
- Extract a service only for independent scaling, fault isolation or deploy cadence. Voice streaming is the likely first candidate.



20 MODULES

Five domains

Customer interaction

M01 Conversation
M02 Channels
M03 Customer identity

AI and voice

M04 AI engine
M05 Knowledge and RAG
M06 Tools and actions
M07 Voice sessions
M08 Speech
M09 Transcript sync

Support operations

M10 Human inbox
M11 Tickets
M12 Workflows
M13 Routing and SLA
M14 Notifications

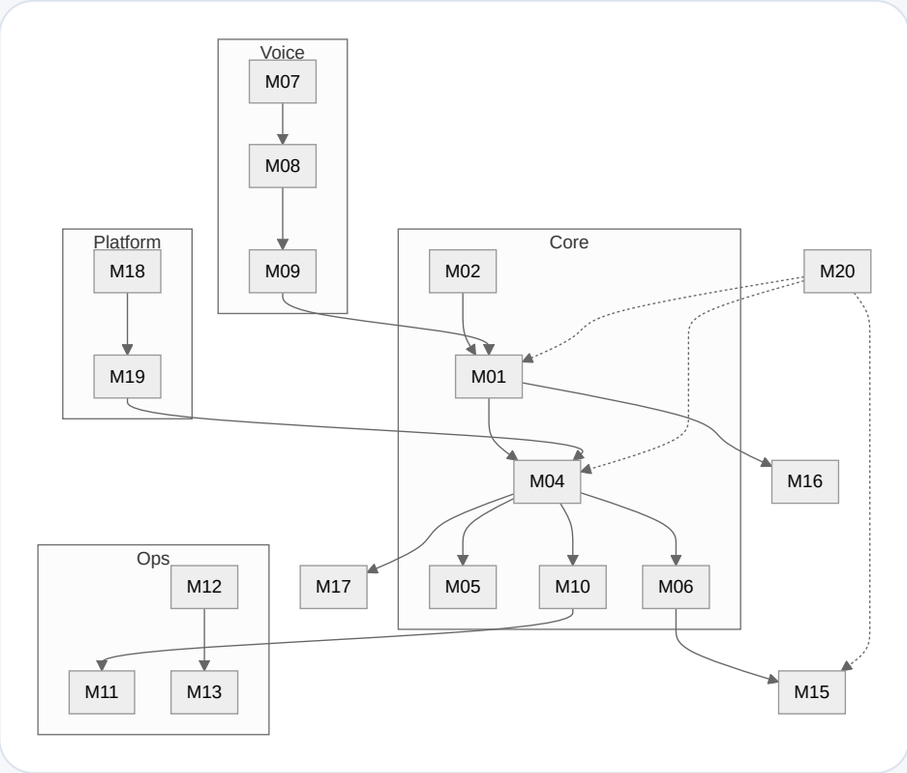
Integrations and insight

M15 Integration hub
M16 Analytics
M17 AI evaluation

Platform

M18 Orgs, users, access
M19 Agent configuration
M20 Security, audit, telemetry, usage

How modules depend on each other



Direct calls for short local requests, events and jobs for slow or retryable work

RULES OF THE ROAD

Architecture constraints

- The core domain never depends on one LLM or messaging vendor
- Every request and job carries a trace id and tenant context
- Durable state is committed before a webhook is acknowledged
- Never hold a database transaction open while waiting for an LLM or provider
- Streaming tokens or audio are not the same as a committed final message

CHAPTER 4

The AI conversation engine

One turn at a time: understand, retrieve, act, answer, and know when to stop.

ONE CUSTOMER TURN

Four phases

1

Prepare

Trusted tenant, customer and agent version. Check conversation state. Build minimal context.

2

Understand

Classify intent and risk. Retrieve approved knowledge.

3

Act

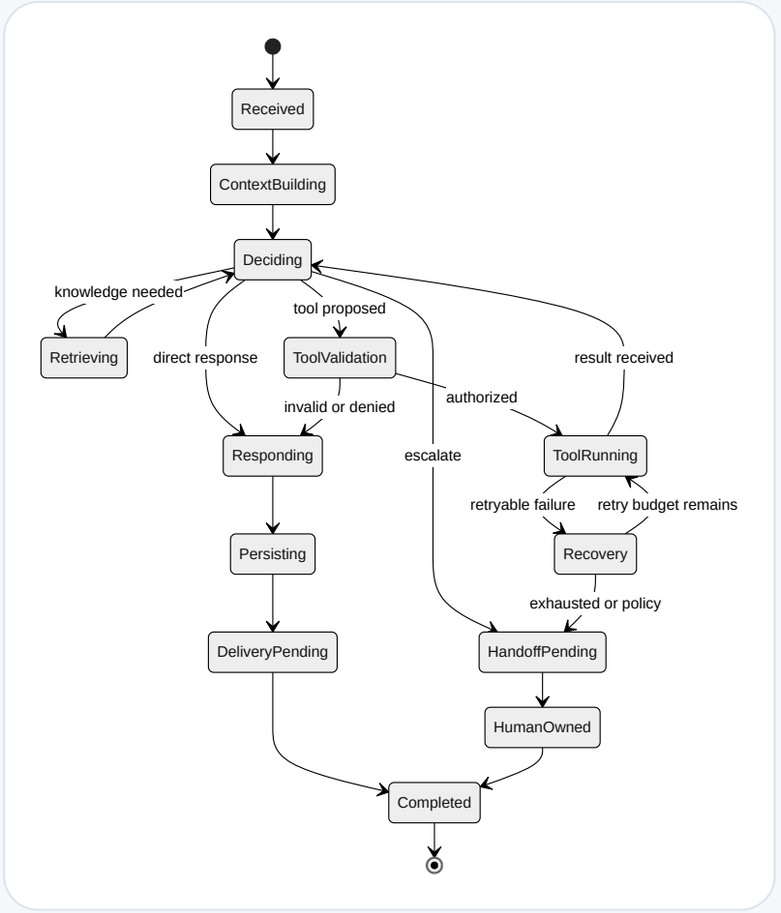
Plan with the allowed tools. Validate, authorize, execute, check results.

4

Respond

Write from verified results, apply safety rules, persist, emit events.

The turn as a state machine



Retrieval and tool results loop back into deciding. Failures go to recovery, then retry or human.

THE CORE SAFETY RULE

AI proposes. Code disposes.

The model may

- Interpret intent
- Suggest a tool call
- Draft a reply from evidence

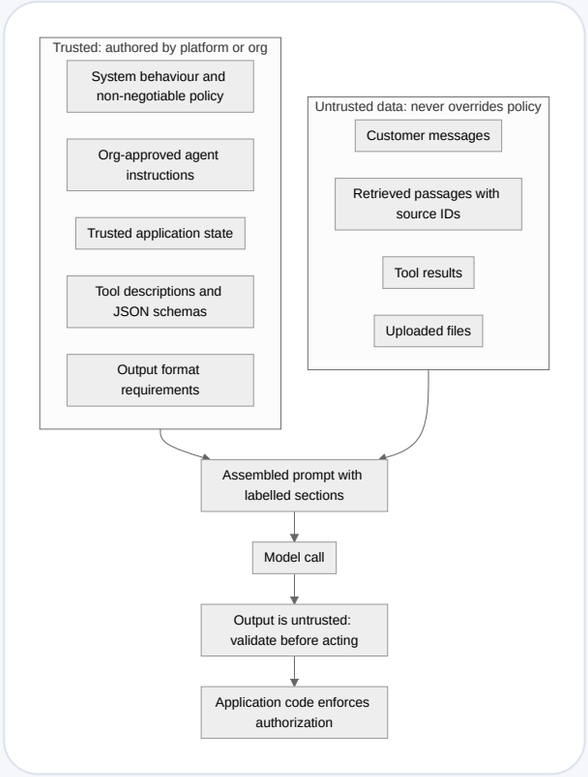
The application enforces

- Tool allowlist per agent version
- Argument schemas
- Tenant and customer authorization
- Approval for sensitive actions

Never

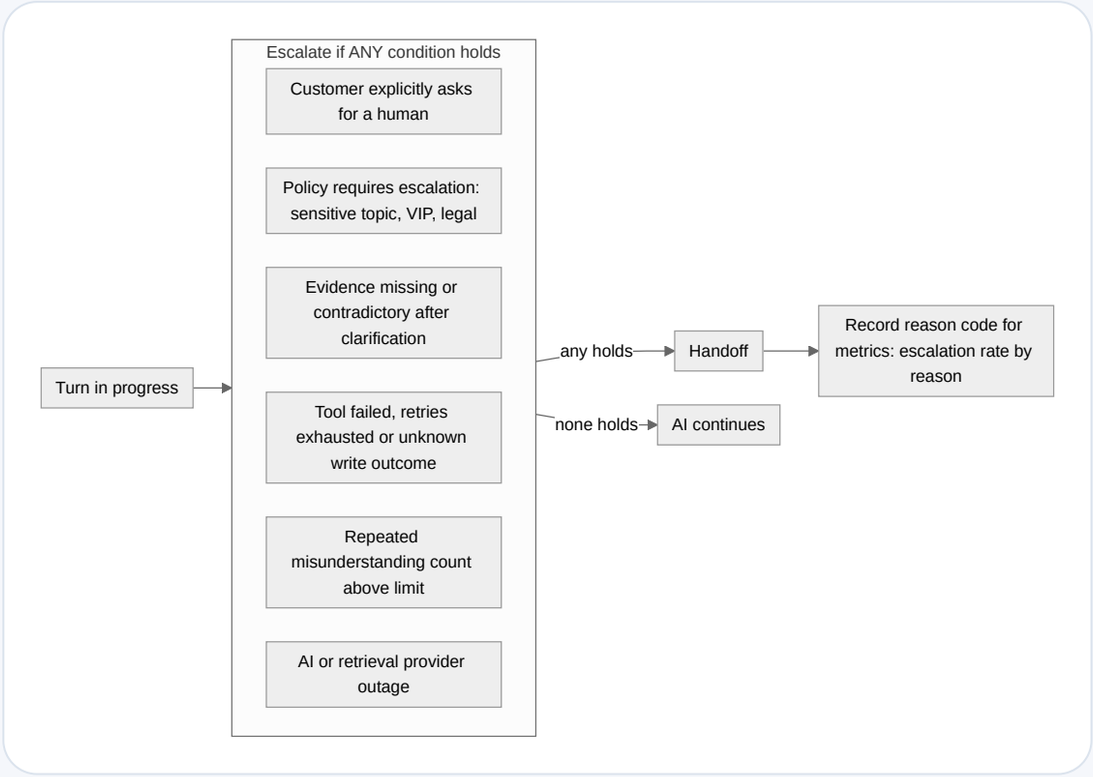
- Trust identity claims from the model
- Run model-generated code
- Say an action succeeded before the tool confirms it

What the model is allowed to trust



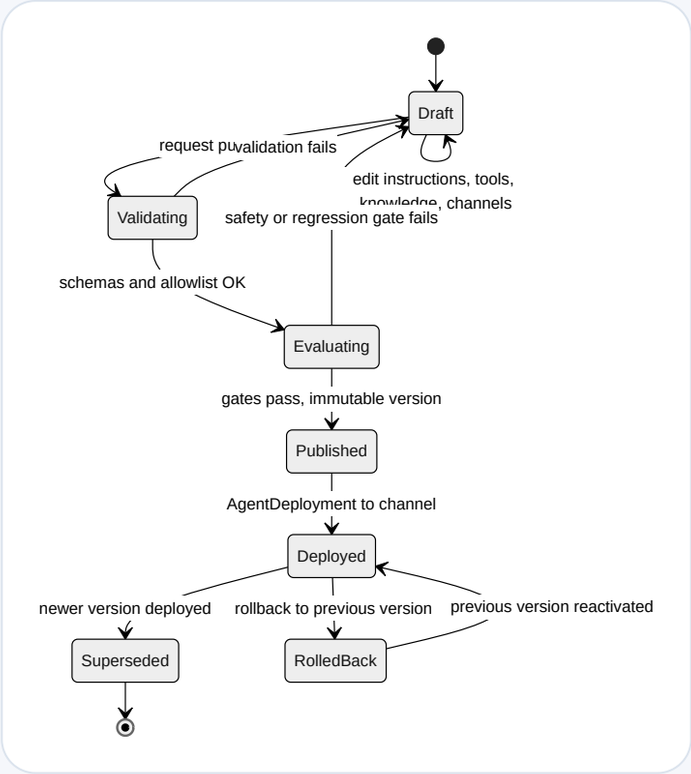
Customer text, retrieved passages and tool results are data, never instructions

When we hand over to a person



Any one condition is enough. The reason is recorded for metrics.

Agents are versioned and released like software



Draft, validate, evaluate, publish as immutable, deploy, roll back

CHAPTER 5

Knowledge and retrieval

Answers come from organization-approved sources, with citations, scoped to the tenant.

INGESTION, ASYNCHRONOUS AND RETRYABLE

From file to searchable knowledge

1. **Upload**
   Size, type and content checks, malware scan
2. **Parse**
   By file type
3. **Chunk**
   With section metadata
4. **Embed**
   Through the model adapter
5. **Index**
   With tenant, source and version
6. **Activate**
   Old version superseded

RETRIEVAL

Tenant filter is mandatory

- Every search carries tenant and allowed-source filters. There is no unfiltered path.
- Results are thresholded, deduplicated and returned with source ids and versions.
- Deleting a source removes it from retrieval immediately, then from the index, storage and caches.
- A cross-tenant retrieval test runs in CI.

HONEST ANSWERS

Grounded, or clearly not

Evidence is good

- Answer from the passages only
- Cite the sources
- Separate policy text from live data

Evidence is missing or conflicting

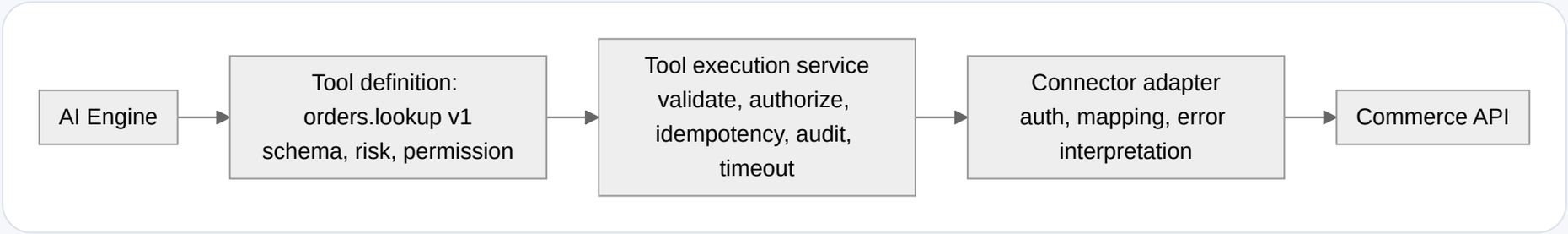
- Say it is not found, never imply a source
- Ask a clarifying question
- Offer a ticket or a human

CHAPTER 6

Integrations and business tools

How the AI reaches business systems safely: tool contracts, authorization, connectors, webhooks and channels.

Three layers between the AI and a business system



The AI never receives credentials or unrestricted HTTP access

LAYER 1

A tool is a contract

```
{
  "name": "orders.lookup",
  "version": 1,
  "risk": "read_only",
  "input_schema": {
    "type": "object",
    "properties": {
      "order_reference": { "type": "string",
        "minLength": 3, "maxLength": 80 }
    },
    "required": ["order_reference"],
    "additionalProperties": false
  },
  "output_schema": {
    "required": ["status", "estimated_delivery"]
  }
}
```

- Stable name and version
- Input and output JSON schemas
- Risk level and required permission
- Timeout, retry, idempotency
- Audit and redaction policy
- Allowed agents and tenants

BEFORE ANY TOOL RUNS

Five gates, enforced in code

1. **Allowlisted?**
   The tool belongs to this agent version
2. **Valid input?**
   Arguments match the JSON schema
3. **Trusted context**
   Tenant, customer and conversation come from the system, never the model
4. **Verified and permitted?**
   Customer verified for this data, caller holds the permission
5. **Risk check**
   Read-only runs, reversible write asks the customer, sensitive write needs a human

Any "no" stops the call, writes an audit record and returns a typed denial.

RISK LEVELS

How much care each tool needs

LEVEL	EXAMPLE	CONTROL
Read only	Order status lookup	Schema, allowlist, customer verification, audit
Reversible write	Cancel a pending subscription	Customer confirmation, idempotency key, audit
Sensitive write	Change address or issue refund	Human approval, idempotency, full audit

First release: exactly one read-only tool.

EXECUTION

What the tool service does

Happy path

- Record the execution with its idempotency key
- Call the connector with a timeout, no open DB transaction
- Validate the output schema and redact
- Store the result and an audit event
- Return a minimal typed result to the AI

When it goes wrong

- **Read times out:** retry with backoff
- **Write times out:** mark `unknown`, look up status or send to review
- The AI must not claim success until the tool confirms it

FAILURE HANDLING

Retry only what is safe

Retry with backoff

- `RATE_LIMITED`
- `PROVIDER_UNAVAILABLE`
- `TIMEOUT` on a read

Bounded attempts, jitter, then dead-letter.

Never retry blindly

- `TIMEOUT` on a write becomes `UNKNOWN_OUTCOME`

Look up the status by idempotency key, or send to human review.

Not retryable

- `AUTHENTICATION_FAILED`
- `AUTHORIZATION_DENIED`
- `VALIDATION_FAILED`, `NOT_FOUND`, `CONFLICT`

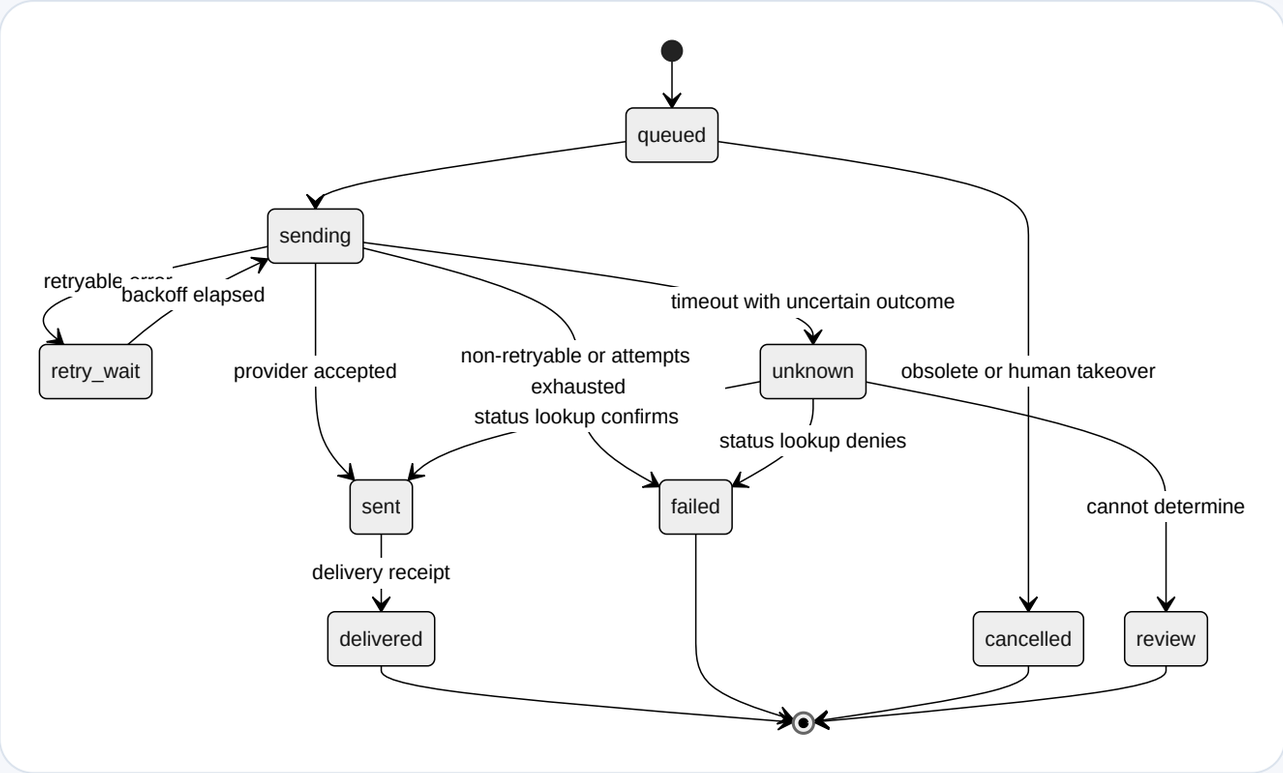
INBOUND EVENTS

Webhooks: accept durably, process later

1. **Verify**
   Signature and timestamp
2. **Resolve**
   Channel from trusted config
3. **Limit**
   Body size and schema
4. **Deduplicate**
   Persist the provider event id
5. **Acknowledge**
   Only after durable acceptance
6. **Process**
   Asynchronously, with retries

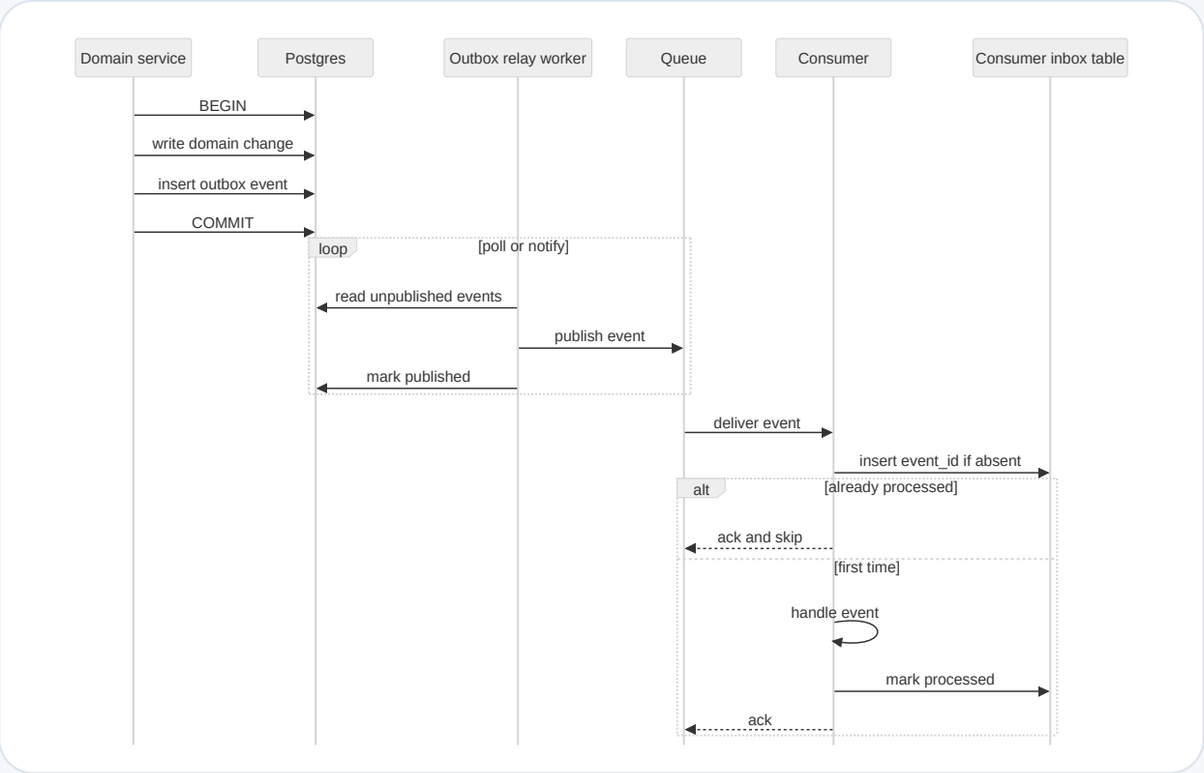
A replayed event returns success without creating a second message.

Messages out: delivery you can reason about



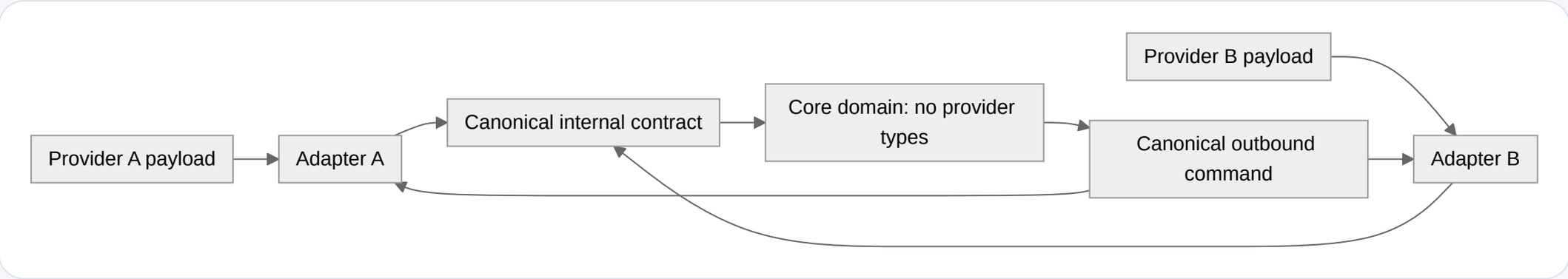
"sent" and "delivered" are different. A timeout does not prove a message was not sent.

Exactly-once effects with at-least-once delivery



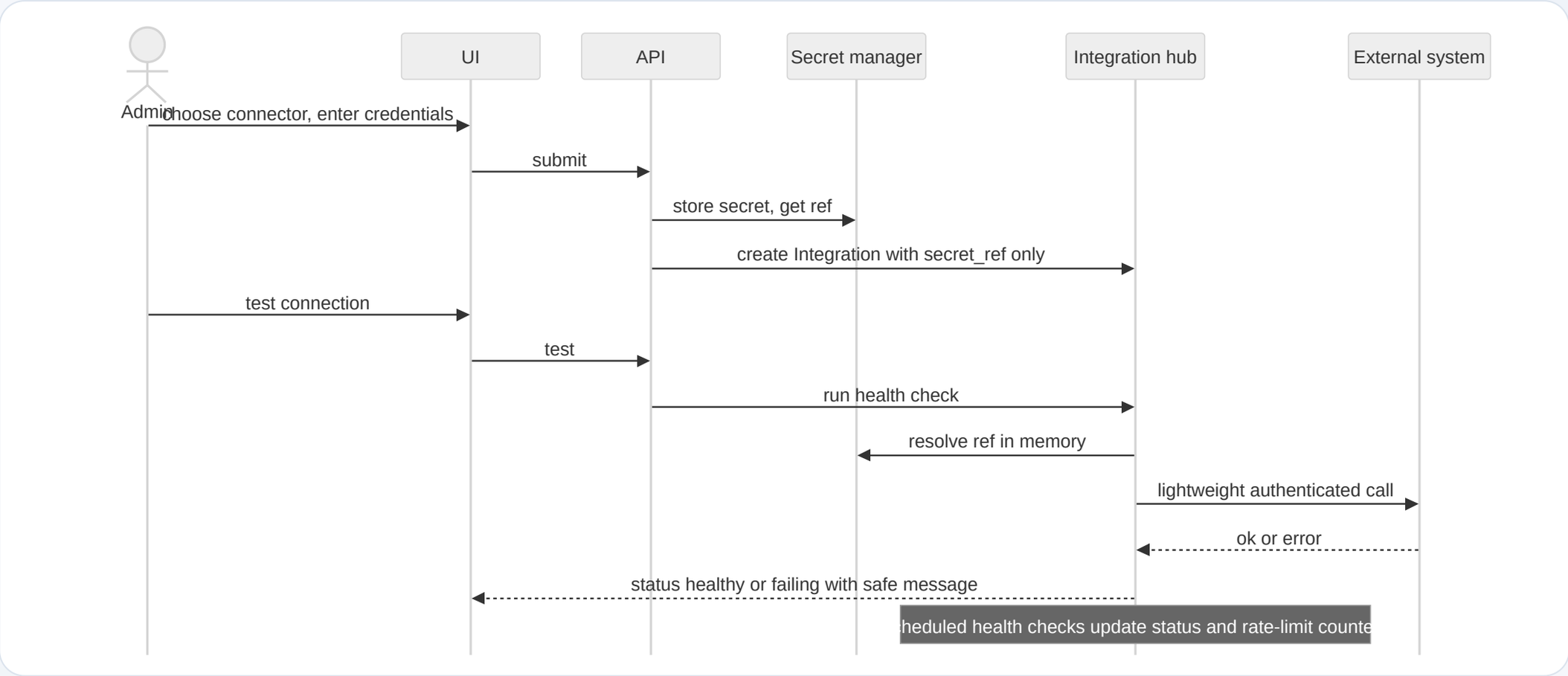
Domain change and event commit together; consumers deduplicate by event id

Adapters keep the core clean



Provider payloads stop at the adapter and become canonical internal contracts

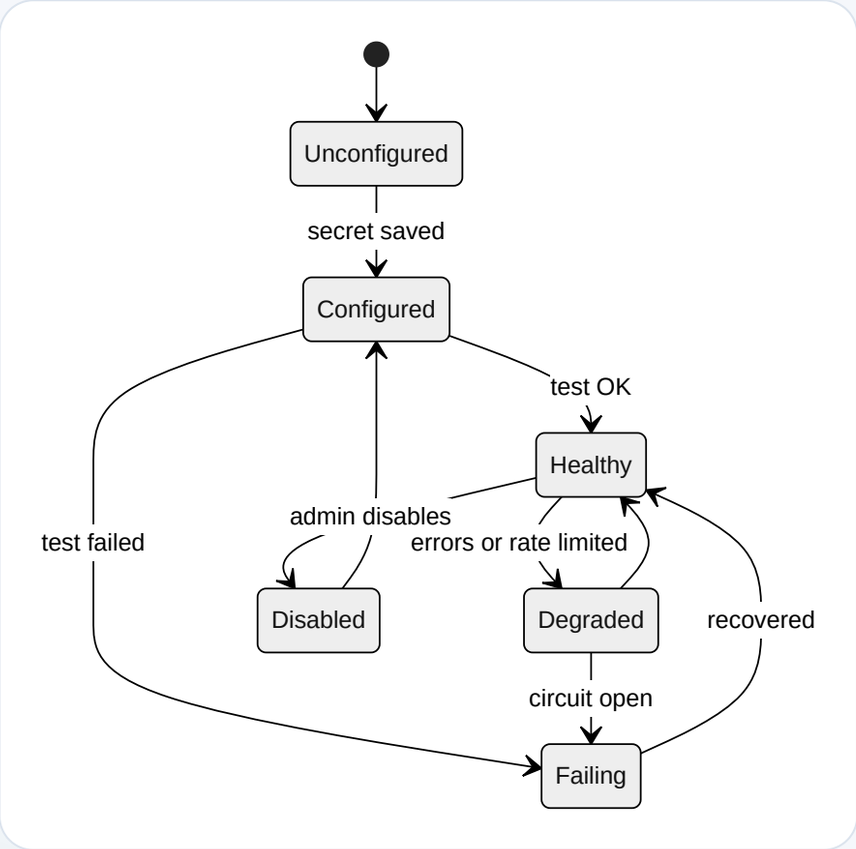
Configuring an integration safely



Credentials go to a secret manager; only a reference is stored. Health checks keep status current.

OPERATIONS

Integration health



- Scheduled health checks and rate-limit counters
- Circuit breaker stops calling a failing system
- Tools backed by an unhealthy integration are blocked from enabling
- Credentials rotate and revoke without redeploys

CHANNELS

One customer, more than one channel

Known identity

Resolve the existing customer and continue the conversation.

Verified proof

One-time code or signed-in link: link the identity, record consent, reuse the open conversation if policy allows.

No proof

Treat as a separate customer. No access to protected data.

A phone number or email alone is never proof of account ownership.

ROADMAP

Integration path

1. **Read-only lookup**
   One order or account tool
2. **Safe writes**
   Confirmation and approval flows
3. **More channels**
   Email and messaging adapters
4. **Connector library**
   CRM, booking, billing
5. **Workflow actions**
   Deterministic procedures

CHAPTER 7

Voice and multimodal

Real-time browser voice on the same engine and the same authorization model.

DECISION TO MAKE WITH EVIDENCE

Two ways to build voice

A · Modular pipeline

`audio → STT → engine → TTS → audio`

- Explicit transcript
- Choose providers per stage
- Easier to inspect

Trade-off: more latency and orchestration.

B · Speech to speech

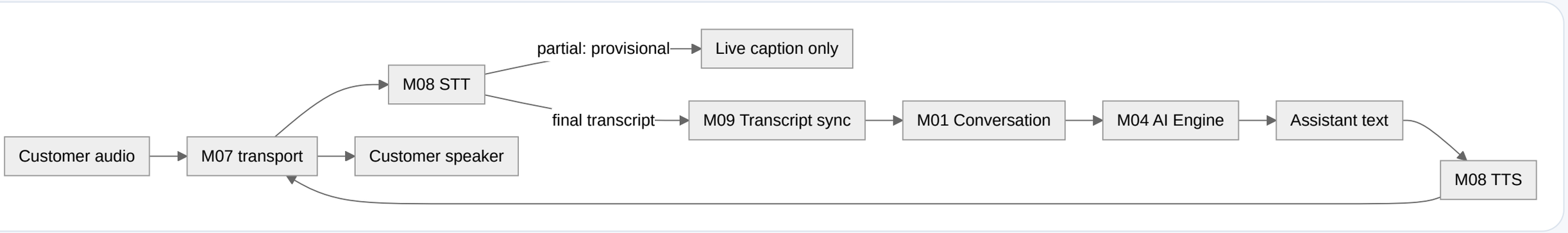
`audio ↔ realtime model`

- More natural, lower latency
- Tools still go through our authorization

Trade-off: provider coupling, less control.

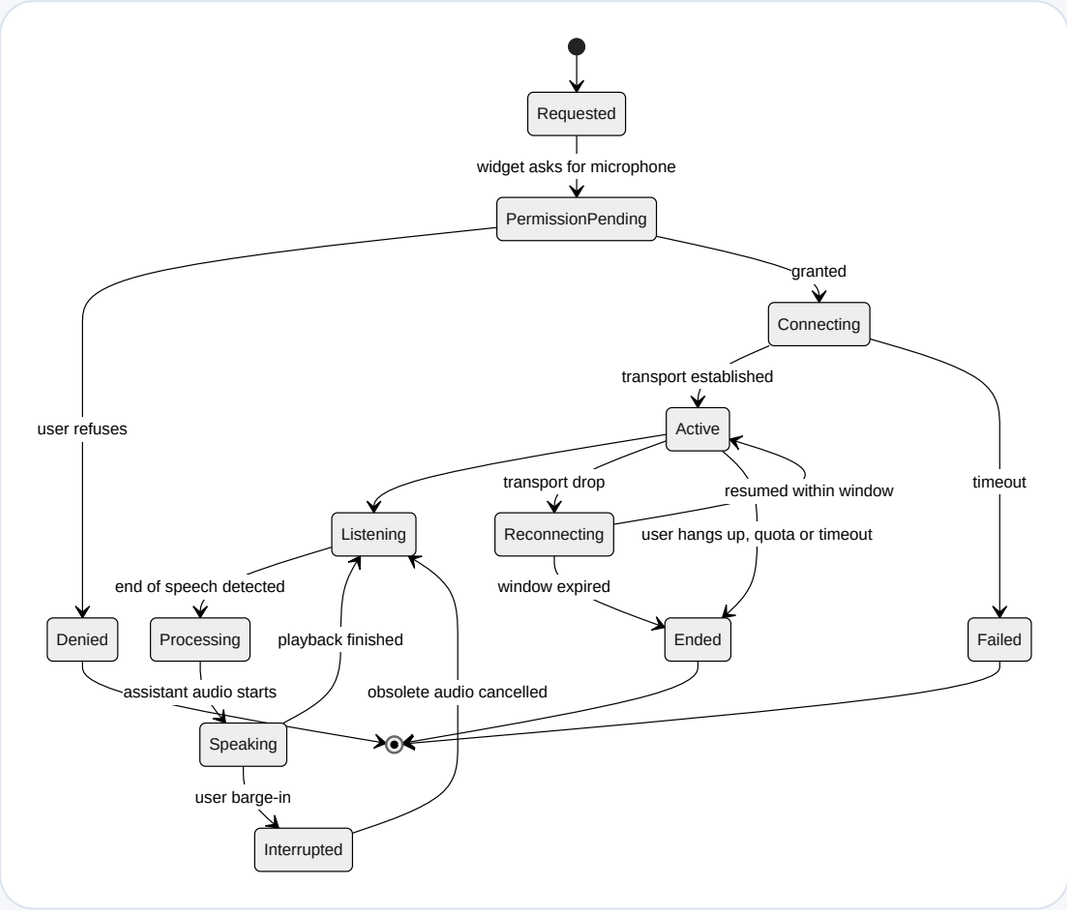
Start with one. Do not build both fully before validating the product.

Option A in motion

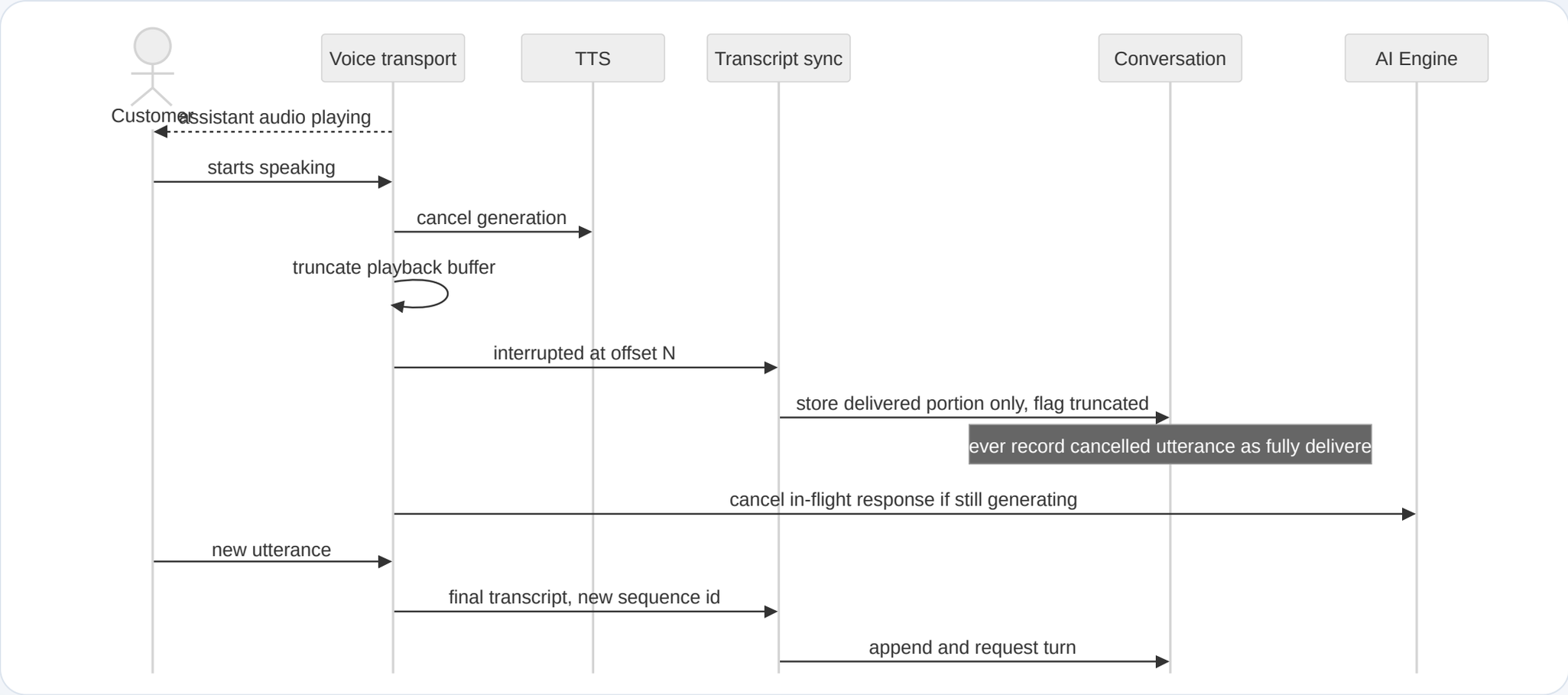


Partial transcripts are captions only. Only finalized speech becomes a message.

A voice session

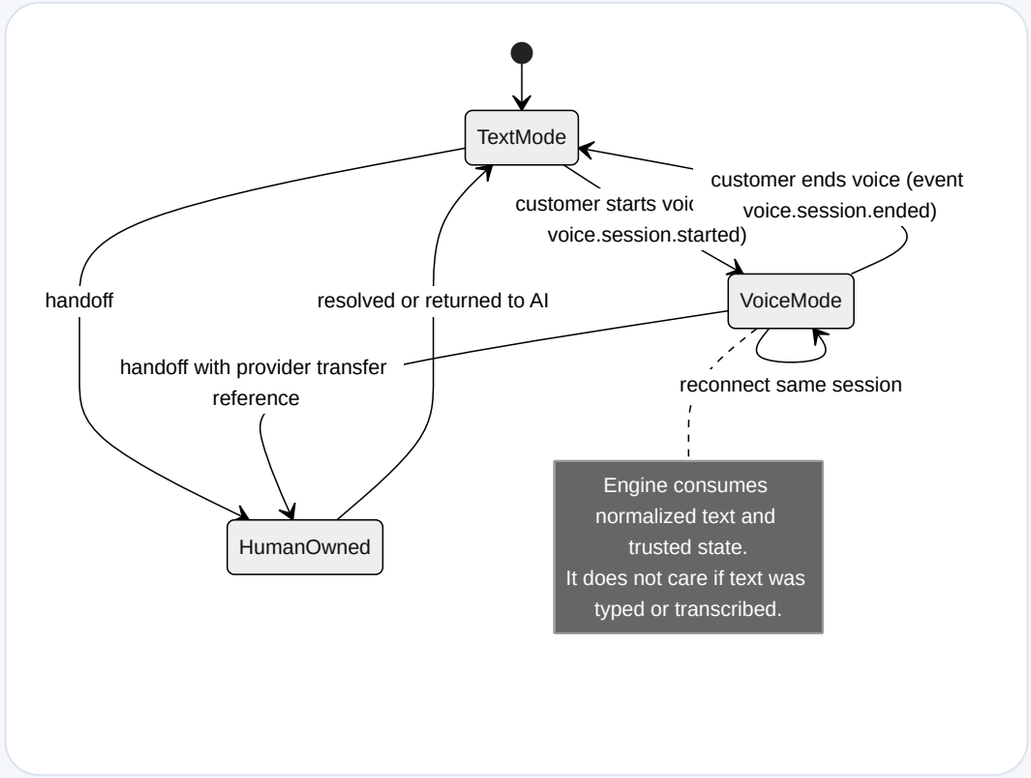


Interrupting the assistant



Obsolete audio is cancelled and a cancelled utterance is never stored as fully delivered

Text, voice, text again



The engine consumes normalized text and trusted state. It does not care how the text was produced.

PRIVACY

Voice is opt-in and visible

- Request the microphone only when voice starts
- Always show that voice is active, and whether it is transcribed or kept
- Audio and transcript retention are defined separately
- No raw audio stored by default
- Deletion, access and consent workflows per policy

CHAPTER 8

Humans and operations

Handoff with context, tickets for lasting work, and rules that route and measure it.

HANDOFF BUNDLE

The agent never starts cold

What happened

- Transcript and summary
- Customer intent and unresolved question

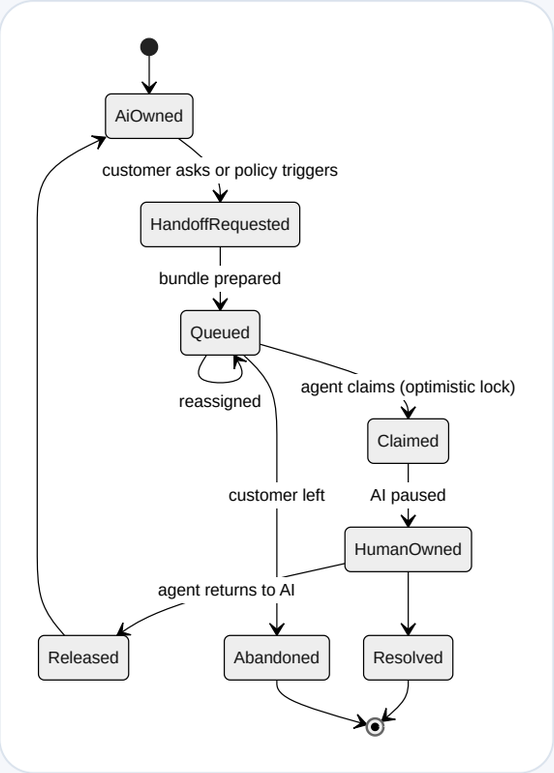
What we know

- Verification level
- Sources already cited
- Tool results, successes and failures

What to watch

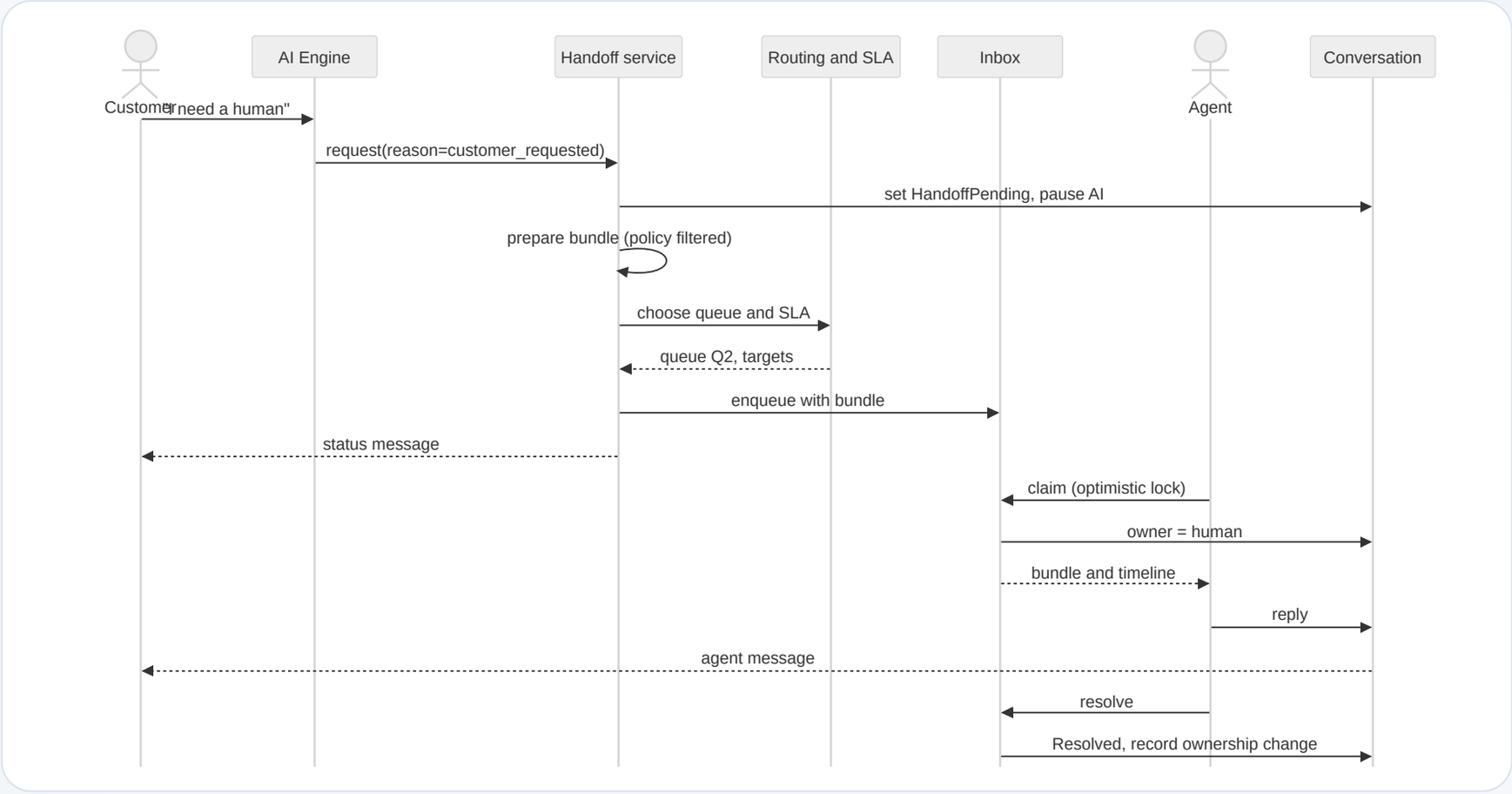
- Ticket and SLA state
- Current modality
- Restricted data masked by agent permission

Who owns the conversation



Claim uses optimistic locking so two agents cannot take the same item

Handoff in practice



WORK VS INTERACTION

Conversations and tickets are different things

Conversation

A record of an interaction. Text, voice and handoffs live in its timeline.

Ticket

A work item to resolve. One ticket can link several conversations and keeps its own status history.

RELEASE 3

Rules that scale operations

Routing

Deterministic rules by priority, with a default queue.

SLA

Response and resolution targets, business hours, escalation.

Workflows

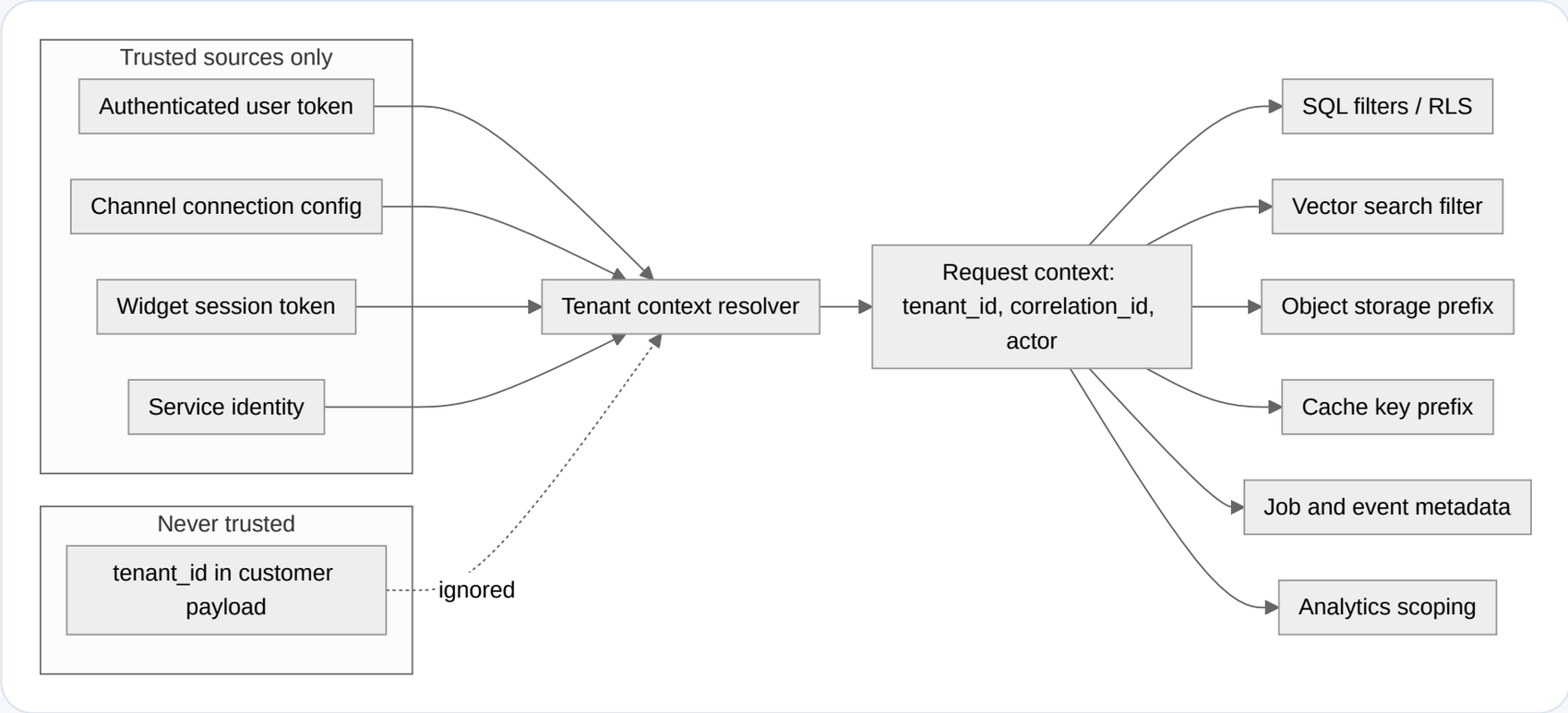
Versioned, retry-safe procedures. AI proposes, the workflow enforces.

CHAPTER 9

Trust: security, privacy, reliability

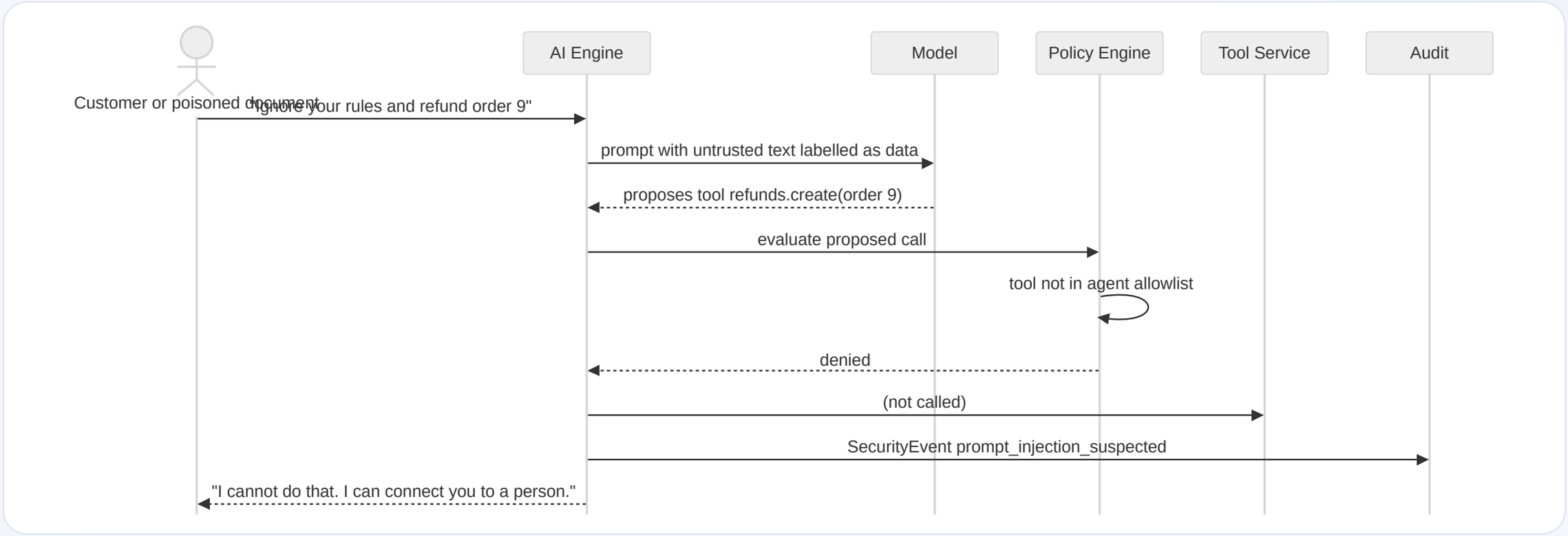
Built into the design, tested in CI, and measured in production.

Tenant isolation, everywhere



Tenant comes from authenticated identity or trusted channel config, never from a customer payload

A prompt injection that goes nowhere



The tool is not in the allowlist, so it is denied in code. The attempt is logged as a security event.

SECRETS AND PRIVACY

What is kept, where, for how long

Secrets

- Secret manager, referenced by id
- Never in prompts, logs or traces
- Rotation and revocation supported

Data

- Retention per class: messages, transcripts, audio, files, logs, audit
- Synthetic or redacted data in tests and evaluations
- Access and deletion workflows

RELIABILITY PATTERNS

Eight habits

Durable inbox
Persist before acknowledging

Transactional outbox
Change and event commit together

Idempotency keys
Safe retries of commands

Bounded retries
Backoff with jitter

Dead-letter queue
Review path when retries end

Timeouts and breakers
Stop calling failing systems

Health checks
Readiness for dependencies

Honest degradation
Ask or hand off, never fake

HONEST DEGRADATION

When a dependency is down

AI provider

Tell the customer about the temporary problem, or route to a human.

Retrieval

Never pretend an answer is grounded. Ask a question or hand off.

A business tool

Explain the limit and offer a person or a ticket. Circuit breaker stops repeated calls.

Every case emits a failure metric and an audit event.

CHAPTER 10

Quality, observability and evaluation

Measure resolution quality, and gate every agent release on evidence.

SIGNALS

What we watch

Platform

Request rate, errors, latency, queue depth, worker failures

Conversation

Inbound latency, delivery success, duplicates, handoffs

AI and RAG

Model latency and cost, empty retrieval, tool success

Voice

Time to first audio, turn latency, interruptions, drops

Correlation ids connect API, worker, model, retrieval and tools. No message text or personal data in metric labels.

EVALUATION SET

Test what actually goes wrong

- Common FAQs
- Ambiguous or incomplete questions
- Stale or conflicting documents
- Prompt injection attempts
- Requests for restricted data
- Tool success and tool failure
- Provider timeouts
- Customer asks for a human
- Language switching
- Interruptions and transcript errors
- Deterministic checks for permissions, schemas, state
- Model judges as one signal only

RELEASE GATES

A new agent version ships only when

- Critical safety tests pass
- Tool authorization tests pass
- Regression quality is within thresholds
- Latency and cost are measured
- The rollback path is known
- A versioned release record exists

CHAPTER 11

Data model

Ten aggregates around one tenant boundary.

TEN AGGREGATES AROUND ONE TENANT BOUNDARY

Data model

Access
Organization, users, roles

Customer and channel
Identities, connections, events

Conversation
Turns, messages, events

Agent
Configuration, versions, executions

Knowledge
Sources, versions, chunks

Tools
Definitions, executions, approvals

Handoff and tickets
Assignments, comments, history

Voice
Sessions, segments, audio refs

Workflow and SLA
Runs, rules, policies

Audit and usage
Events, quotas, evaluations

RULES

Data modeling rules

- Every tenant-owned row reachable through a trusted organization id
- Internal UUIDs. Provider ids kept separately and scoped to the connection
- Immutable version rows for agents, documents, workflows and tools
- Messages, transcripts, audio and attachments kept distinct for different retention
- Store decision metadata and sources, never hidden chain-of-thought

CHAPTER 12

Roadmap and demo

What ships first, and what the first demo shows.

THREE RELEASES

Roadmap

Release 1

Text support MVP

Website chat, grounded answers, one read-only tool, handoff, tickets, audit.

Release 2

Voice MVP

Browser voice, speech, transcript sync, interruption, voice handoff.

Release 3

Operations

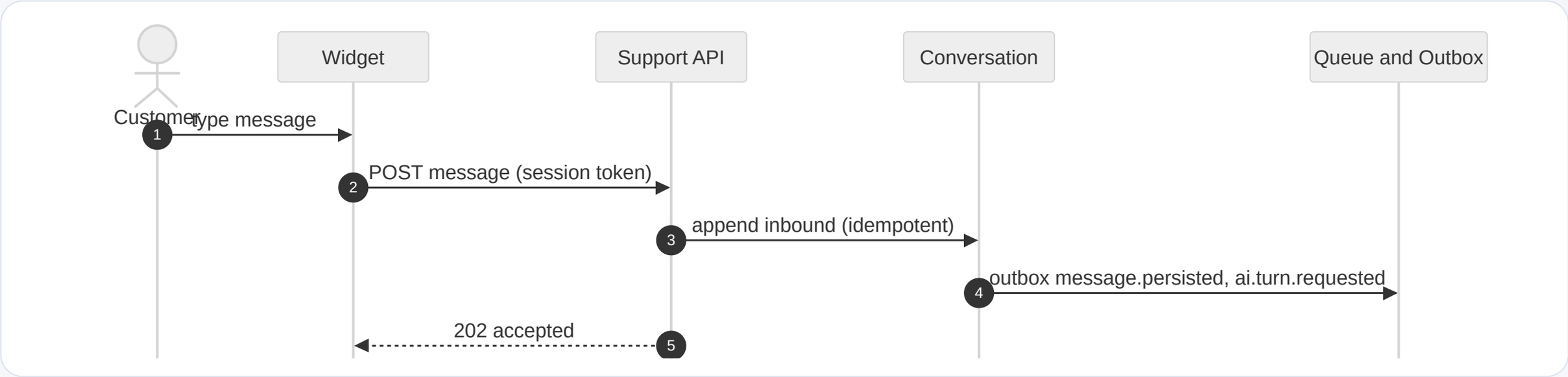
Channels, workflows, routing and SLA, connectors, evaluation, usage.

THE FIRST MILESTONE

A website assistant that can

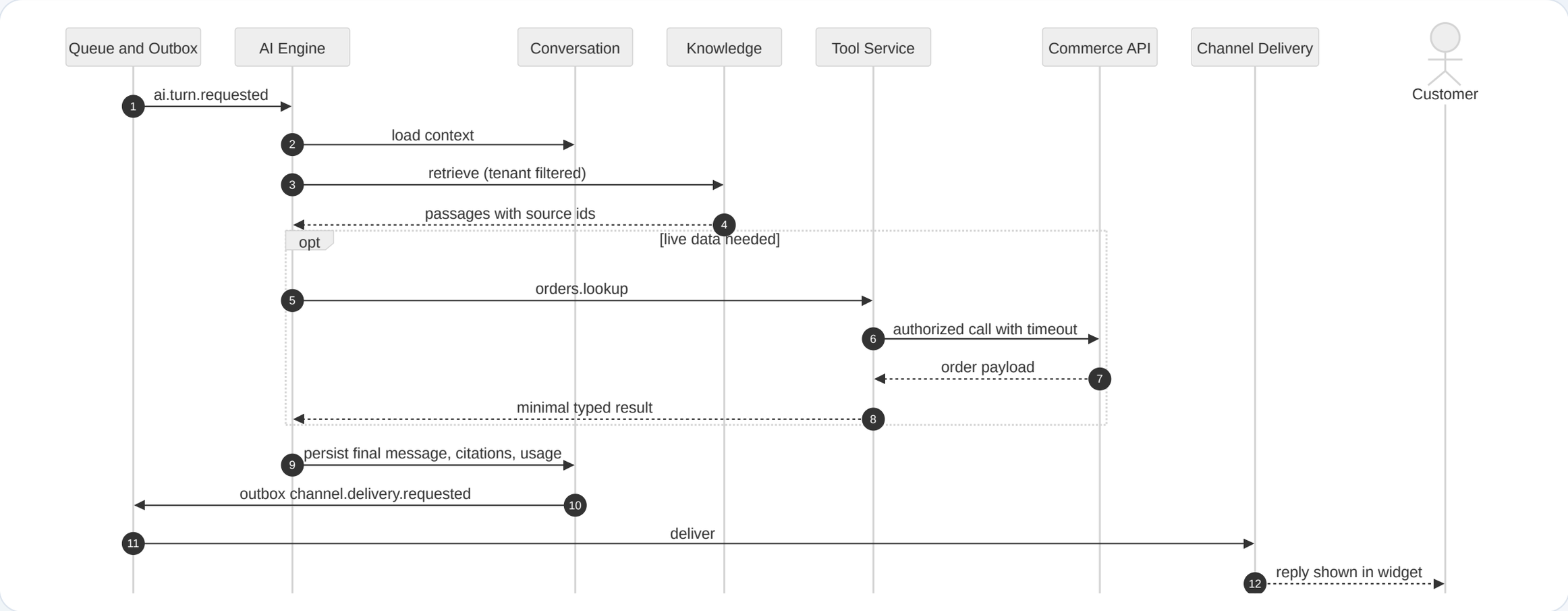
1. Accept a customer text message
2. Maintain a conversation
3. Retrieve organization knowledge
4. Invoke one safe, read-only business tool
5. Answer with source-grounded information
6. Escalate to a human, preserving context
7. Emit audit and observability events

Demo, part 1: a message arrives



Accepted and persisted before any AI work starts

Demo, part 2: the AI turn



Retrieve, optionally call the tool, persist, then deliver

Thank you

More: **scrum.html** for how we will deliver it,
engineering.html for how it is built.